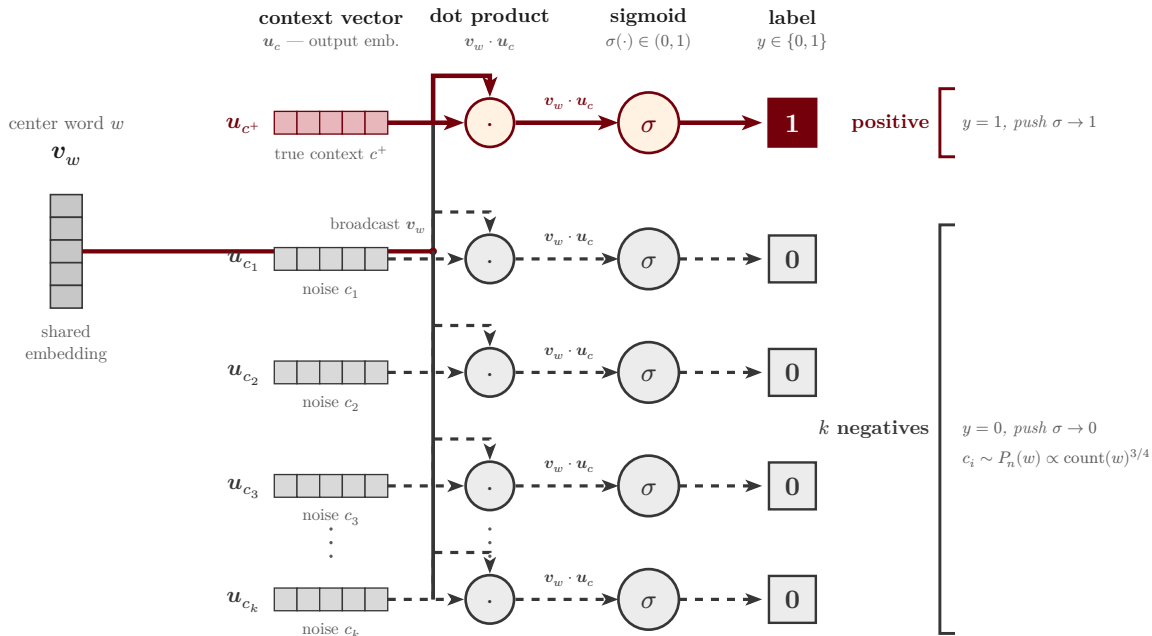


# Skip-gram with negative sampling

binary logistic classifier over word-pairs — not a full softmax



$$\mathcal{L} = -\log \sigma(v_w \cdot u_{c^+}) - \sum_{i=1}^k \log \sigma(-v_w \cdot u_{c_i})$$

maximise the score of the true (center, context) pair; minimise it for  $k$  sampled noise pairs